

Lecture: 12 Successive Differentiation

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Trigonometrical Transformation

It is possible to break up products of powers of sines and cosines into a sum by trigonometry.

Example. If $y = \sin^2 x \cos^2 x$, then find y_n .

Solution. We have, $y = \sin^2 x \cos^2 x = \frac{1}{4}(2 \sin x \cos x)^2 = \frac{1}{4} \sin^2 2x = \frac{1}{4} \left(\frac{1 - \cos 4x}{2} \right)$
i.e.

$$y = \frac{1}{8} - \frac{1}{8} \cos 4x.$$

$$\Rightarrow y_n = 0 - \frac{1}{8} 4^n \cos \left(4x + \frac{n\pi}{2} \right)$$

{Using the previous result $D^n \cos(ax + b) = a^n \cos(ax + b + n\pi/2)$ }

$$\text{i.e., } y_n = -2^{n-2} \cos \left(4x + \frac{n\pi}{2} \right).$$

Example. If $y = \tan^{-1} \frac{(1+x)}{(1-x)}$, then find y_n .

Solution. We have

$$y = \tan^{-1} \frac{(1+x)}{(1-x)} = \tan^{-1} 1 + \tan^{-1} x$$

$$\Rightarrow y_1 = \frac{1}{1+x^2} = \frac{1}{2i} \left(\frac{1}{x-i} - \frac{1}{x+i} \right)$$

$$\Rightarrow y_n = \frac{(-1)^{n-1}(n-1)!}{2i} \left(\frac{1}{(x-i)^n} - \frac{1}{(x+i)^n} \right).$$

Put $x = r \cos \theta$, $1 = r \sin \theta$, where $r = \sqrt{1+x^2}$ and $\theta = \tan^{-1} \left(\frac{1}{x} \right)$.

Then,

$$x + i = r(\cos \theta + i \sin \theta) \Rightarrow (x + i)^n = r^n(\cos n\theta + i \sin n\theta)$$

$$\Rightarrow \frac{1}{(x+i)^n} = r^{-n}(\cos n\theta - i \sin n\theta) \text{ and } \frac{1}{(x-i)^n} = r^{-n}(\cos n\theta + i \sin n\theta)$$

$$y_n = \frac{(-1)^{n-1}(n-1)!}{2i} \{ r^{-n}(\cos n\theta + i \sin n\theta) - r^{-n}(\cos n\theta - i \sin n\theta) \}$$

$$\Rightarrow y_n = (-1)^{n-1}(n-1)! r^{-n} \sin n\theta$$

as $r^{-n} = \sin^n \theta$

$$\Rightarrow y_n = (-1)^{n-1}(n-1)! \sin^n \theta \sin n\theta$$

Leibnitz's Theorem

This theorem is used for finding the n th differential coefficient of a product. The theorem is as follows:

Theorem 1. *If u and v be any two functions of x , then*

$$D^n(uv) = (D^n u).v + \binom{n}{1} D^{n-1} u.Dv + \binom{n}{2} D^{n-2} u.D^2 v + \dots + \binom{n}{r} D^{n-r} u.D^r v + \dots + u.D^n v. \quad (1)$$

Proof. We shall prove this theorem by mathematical induction. Let $y = uv$, where u and v are functions of x . Differentiating both sides w.r.t. x , we get

$$y_1 = u_1 v + u v_1 = u_1 v + \binom{1}{1} u v_1.$$

This implies that Leibnitz's theorem is true for $n = 1$.

Differentiating y_1 w.r.t. x , we get

$$y_2 = (u_2 v + u_1 v_1) + (u_1 v_1 + u v_2) = u_2 v + 2u_1 v_1 + u v_2.$$

Clearly, y_2 can be written as

$$y_2 = u_2 v + \binom{2}{1} u_1 v_1 + \binom{2}{2} u v_2$$

This implies that, Leibnitz's theorem is true for $n = 2$.

Now, suppose that the above result is true for a particular value of n . Then differentiating with respect to x , we have

$$D^{n+1}(uv) = \{(D^{n+1} u).v + D^n u.Dv\} + \binom{n}{1} \{D^n u.Dv + D^{n-1} u.D^2 v\} + \dots + \binom{n}{r} \{D^{n-r+1} u.D^r v + D^{n-r} u.D^{r+1} v\} + \dots + \{Du.D^n v + u.D^{n+1} v\}.$$

After rearranging this, we get

$$D^{n+1}(uv) = (D^{n+1} u).v + \left(1 + \binom{n}{1}\right) (D^n u.Dv) + \dots + \left(\binom{n}{r} + \binom{n}{r+1}\right) (D^{n-r} u.D^{r+1} v) + \dots + u.D^{n+1} v.$$

But,

$$\binom{n}{r} + \binom{n}{r+1} = \binom{n+1}{r+1}.$$

Hence,

$$D^{n+1}(uv) = (D^{n+1} u).v + \binom{n+1}{1} D^n u.Dv + \dots + \binom{n+1}{r+1} D^{n-r} u.D^{r+1} v + \dots + u.D^{n+1} v.$$

Therefore, if the theorem is true for any value of n , it will also be true for $(n + 1)$. Thus it must be true for every positive integral value of n .

Example. Find the n th differential coefficient of $x^3 \log x$.

Solution. By Leibnitz Theorem, if $y = uv$, then we are given that $y = x^3 \log x$. Therefore, we can take

$$u = \log x, v = x^3.$$

On differentiating u and v successively, we have

$$u_1 = \frac{1}{x}, u_2 = \frac{-1}{x^2}, u_3 = \frac{(-1)(-2)}{x^3}, \dots, u_n = \frac{(-1)^{n-1}(n-1)!}{x^n}$$

$$\text{and } v_1 = 3x^2, v_2 = 6x, v_3 = 6, v_4 = 0, \dots, v_n = 0.$$

Substituting these values in equation (1), we get

$$y_n = \frac{(-1)^{n-1}(n-1)!}{x^n} x^3 + \binom{n}{1} \frac{(-1)^{n-2}(n-2)!}{x^{n-1}} (3x^2) + \binom{n}{2} \frac{(-1)^{n-3}(n-3)!}{x^{n-2}} (6x) \\ + \binom{n}{3} \frac{(-1)^{n-4}(n-4)!}{x^{n-3}} (6) + 0 + 0 + \dots$$

Further, simplify it by yourself.

Example. If $u = \tan^{-1} x$, show that

$$(1 + x^2) \frac{d^2 u}{dx^2} + 2x \frac{du}{dx} = 0;$$

hence determine the values of all the derivative of u with respect to x , when $x = 0$.

Solution Here, $u = \tan^{-1} x$, then $u_1 = \frac{1}{1+x^2}$.

$$\Rightarrow (1 + x^2)u_1 = 1$$

Differentiating this equation again we have

$$(1 + x^2)u_2 + 2xu_1 = 0 \text{ or } (1 + x^2) \frac{d^2 u}{dx^2} + 2x \frac{du}{dx} = 0.$$

Differentiating this n times by Leibnitz's theorem,

$$D^n \{(1 + x^2)u_2\} = (1 + x^2)u_{n+2} + n(2x)u_{n+1} + \frac{n(n-1)}{2}(2)u_n,$$

$$D^n \{2xu_1\} = 2xu_{n+1} + n(2)u_n,$$

Adding,

$$0 = (1 + x^2)u_{n+2} + 2(n+1)xu_{n+1} + n(n+1)u_n.$$

Putting $x = 0$, this gives

$$(y_{n+2})_0 = -n(n+1)(u_n)_0.$$

From above, we have $(u_1)_0 = 1$ and $(u_2)_0 = 0$.

Hence, $u_2, u_4, u_6, \dots$ are all zero at $x = 0$. But if n is odd,

$$\begin{aligned}(u_n)_0 &= -(n-1)(n-2)(u_{n-2})_0 \\ &= (n-1)(n-2)(n-3)(n-4)(u_{n-4})_0 = \text{etc.}, \\ \text{so, } (u_n)_0 &= (-1)^{(n-1)/2}(n-1)!. \end{aligned}$$

Hence, $(y_n)_0 = 0$ when n is even, and $(-1)^{(n-1)/2}(n-1)!$ when n is odd.

Reference Book: Differential Calculus, Dr. Gorakh Prasad. Pothishala Private LTD.